AQUAREV WATER · COMMERCIAL

Pump Room Implementation Guide

How to assess, approve, and install an AquaRev device on a hotel or resort pool system, with the paperwork a municipality needs to confirm that no permit is required.

Revision 1.1
September 2026

Companion to
Product Specification Guide 2026
Technical White Paper Rev 1.0

aquarevwater.us/data

1 What's in this guide

An AquaRev installation is a half-day plumbing task per body of water. Almost all of the effort is in preparation: confirming the pipe, the flow, the space, and the paperwork before anyone cuts anything. Each section below is marked with who needs to read it.

2 **What is being installed** — MANAGEMENT · CHIEF ENGINEER · CONTRACTOR · AHJ

3 **Site assessment** — CHIEF ENGINEER · CONTRACTOR

4 **Sizing and specifications** — CHIEF ENGINEER · CONTRACTOR · AHJ

5 **Approvals** — MANAGEMENT · CHIEF ENGINEER

6 **Permit determination** — MANAGEMENT · CHIEF ENGINEER · AHJ

7 **Jurisdiction package** — CHIEF ENGINEER · AHJ

8 **Installation** — CONTRACTOR · CHIEF ENGINEER

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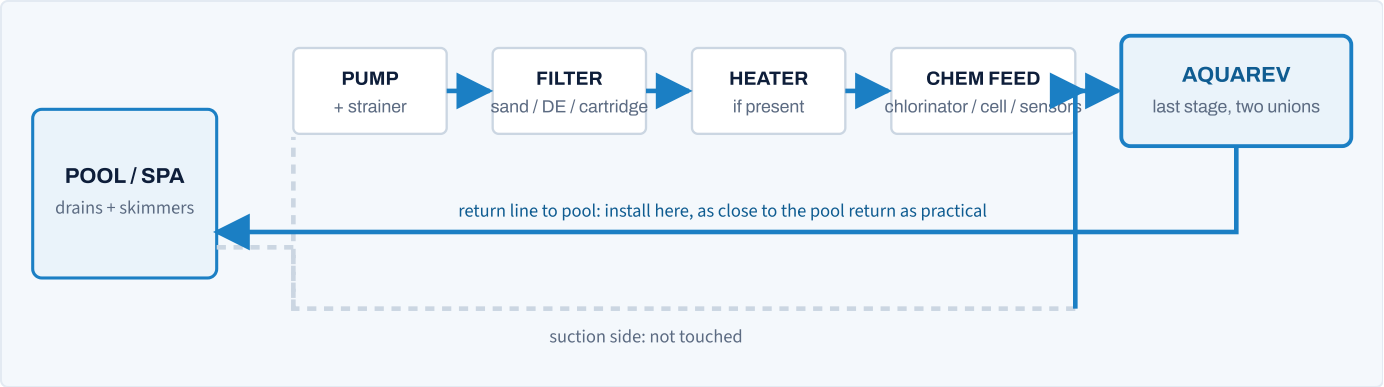
THE ONE RULE THAT NEVER CHANGES

The AquaRev device supplements the pool's disinfection system. It does not replace it. Free chlorine and pH must still be held inside the regulated range, tested on schedule, and recorded, exactly as before.

2 What is being installed

MANAGEMENT | CHIEF ENGINEER | CONTRACTOR | AHJ

A passive PVC device that mounts in-line on the pool return pipe, after the pump, filter, heater, and chemical feeders. Water flowing through its fixed internal shape undergoes hydrodynamic cavitation. There is nothing to power, dose, clean, calibrate, or replace.



The device is always the last component before water re-enters the pool, so every sensor, feeder, and controller upstream sees water exactly as it did before. Suction-side plumbing and drain covers are not touched.

ATTRIBUTE	FACT	SOURCE
Listing	Water Conditioning Device, certified to NSF/ANSI/CAN 50-2024 by IAPMO R&T. Bears the USPC® mark. Listed in compliance with the Uniform Swimming Pool, Spa and Hot Tub Code.	IAPMO R&T Listing, File No. W-12547, current as of 4/30/2026
Materials	PVC body and fittings. NSF/ANSI 372 lead-free. California Prop 65 compliant. Made in the USA.	Spec Guide 2026; NSF binder
Moving parts, power, consumables	None. No filter media, screens, lamps, cartridges, or chemicals.	Spec Guide 2026
Connections	In-series on the existing return pipe, using two supplied Schedule 40 PVC unions. No bypass. No electrical, gas, water-supply, drain, or sewer connection.	White Paper §6.3

ATTRIBUTE	FACT	SOURCE
Existing equipment	Pump, filter, heater, chlorinator or salt cell, controller, and sensors stay in place, upstream, unmodified.	White Paper §6.1
Head loss	2.6 psi rated. IAPMO measured 0.4 psi at 26 gpm up to 2.6 psi at 72 gpm on the tested unit.	Spec Guide; Listing footnote 2
Certified claim	41% less chlorine needed to maintain the test pool's water chemistry.	Listing footnote 1
Warranty	Lifetime limited. Requires professional installation and balanced water chemistry.	aquarevwater.us/warranty

3 Site assessment

CHIEF ENGINEER | CONTRACTOR

One assessment per body of water. Each recirculating loop gets its own device. Walk the pump room with a tape measure and a camera and fill in one record per loop.

Pump-room walk

- [ ] **Body of water**
  Name, volume in gallons, pool or spa, indoor or outdoor.
- [ ] **Return pipe size and material**
  Measure the outside diameter. Confirm PVC. Other materials need transition fittings.
- [ ] **Flow rate in gpm**
  From the flow meter or the pump curve. It must fall inside the model's range in Section 4.
- [ ] **Pump pressure margin**
  Read the filter gauge with a clean filter. The device adds up to 2.6 psi.
- [ ] **Last treatment component**
  Find the heater outlet, chlorinator or salt cell, and chemical injection points. The device goes after all of them.
- [ ] **Space on the return line**
  Straight run of at least the device length plus 4 in. each side, and clearance for the body diameter. Any orientation works. If short, plan a reroute with elbows.
- [ ] **Isolation and drain-down**
  Are there valves to isolate the return line? If not, plan to drain the filter and refill.
- [ ] **Photos**
  Wide shot of the equipment, close-up of the cut zone with a tape in frame, gauge readings.

SEND THE RECORD TO AQUAREV BEFORE ORDERING WHEN

flow is within 10% of the model's minimum or maximum, the pool exceeds 500,000 gallons, or the return pipe is 8 in. or larger. Loops above a model's maximum can be split across two devices in parallel.

Assessment record

PROPERTY / BODY OF WATER	
VOLUME (GAL)	
RETURN PIPE OD / MATERIAL	
FLOW (GPM) AND HOW MEASURED	
FILTER GAUGE, CLEAN (PSI)	
STRAIGHT RUN AVAILABLE (IN.)	
ISOLATION VALVES / DRAIN-DOWN	
MODEL SELECTED	
ASSESSED BY / DATE	

4 Sizing and specifications

CHIEF ENGINEER | CONTRACTOR | AHJ

Select by return pipe size, then confirm the loop's flow sits inside that model's range. Figures are from the Product Specification Guide 2026. Eight-inch and ten-inch units are available on request.

RETURN PIPE	2 IN.	3 IN.	4 IN.	6 IN.
Product ID	KD2X4X2	KD3X6X3	KD4X8X4	KD6X12X6
Flow range	44–127 gpm	112–273 gpm	240–480 gpm	440–1,100 gpm
Head loss	2.6 psi	2.6 psi	2.6 psi	2.6 psi
Connection	2 in. PVC hub	3 in. PVC hub	4 in. PVC hub	6 in. PVC hub
Body diameter (D1)	5.0 in.	7.25 in.	9.34 in.	13.75 in.
Overall length (L1)	20.25 in.	24.0 in.	28.19 in.	40.0 in.
Shipping weight	8 lb	20 lb	35 lb	50 lb
Unions supplied	2 × Sch 40	2 × Sch 40	2 × Sch 40	2 × Sch 40

CUT LENGTH

The pipe section removed is shorter than the overall length, because the union halves that stay on the pipe take up part of it. Measure the unit in hand, socket bottom to socket bottom, before marking the pipe. Field installations of the 2 in. unit have used a 17.5 in. cut.

5 Approvals

MANAGEMENT | CHIEF ENGINEER

Four approvals sit between a quote and a cut pipe. Run them in parallel. None takes more than a few days when the package in Section 7 is attached.

APPROVAL	WHAT THEY NEED	OWNER
Capital or R&M spend	Quote, ROI calculator output, warranty terms.	GM / Ownership
Brand engineering standards	NSF/ANSI 50 listing and spec sheet. Most branded programs accept NSF-50 listed pool equipment without further review.	Chief Engineer
Permit determination	Section 7 package, sent to the building department and the public-pool health program. One person handles it, one voice.	Chief Engineer or AquaRev rep
Pool closure window	Three to five hours per loop, including cement cure. Schedule with Recreation, post signage.	Chief Engineer

Use a licensed pool contractor or plumber insured to the property's normal vendor requirements. The warranty requires professional installation, and an inspector who asks questions will ask who did the work first.

6 Permit determination

MANAGEMENT | CHIEF ENGINEER | AHJ

Most jurisdictions treat this as minor plumbing work on existing pool equipment that needs no permit. The way to get that answer is to ask in writing, describe the device accurately, and keep the reply. The decision is always the authority's to make.

Why the work usually needs no permit

Codes exempt ordinary repair and in-kind replacement of pipe and components, and public-pool health programs review changes to the pool shell, drains, barriers, turnover design, or disinfection method. An AquaRev cut-in touches none of those:

- **No new utility connection.** No electrical, gas, water-supply, drain, or sewer work.
- **No change to disinfection.** Chlorinator or salt cell, feeders, and controllers stay where they are. Required sanitizer and pH levels are unchanged.
- **No change to hydraulics.** Same pump, filter, and turnover. The device is rated for the loop's flow and adds at most 2.6 psi.
- **No suction-side, structural, or barrier work.** Drains, covers, deck, shell, and fencing are untouched.
- **Listed equipment, standard joints.** An NSF/ANSI 50 certified PVC component joined with Schedule 40 unions and solvent cement.

How to get the determination

01 Identify who decides

Usually two offices: the city or county building department for plumbing permits, and the environmental-health or public-pool program. Ask each front counter who owns "changes to public pool equipment."

02 Call, then email the package the same day

Describe the device as Section 2 does and ask one question: does an in-line, listed, passive PVC water-conditioning device installed on the existing return

line with unions require a permit, plan review, or notification? Note the name and time, then send the Section 7 package so the answer is on the record.

03 Get the answer in writing

An email reply is enough. If a code section is cited, copy it word for word into the determination record. If a permit is required, ask the type, fee, and whether plan check applies.

04 File it before install day

The determination record and email thread go in the project file. Give the contractor a copy to carry on site.

> **DO NOT**
>
> Do not call the device a filter, a chlorinator, or a replacement for either. Do not install ahead of a required permit. Do not ask a second office for a different answer after one has said a permit is required. An accurate description gets a fast "no permit" in most places; an inaccurate one puts the pool's operating permit at risk.

If a permit is required, the Section 7 package is the submittal. Expect an over-the-counter minor plumbing permit with a nominal fee and an inspection that consists of looking at two unions and a listing mark. Ask whether the inspector wants to see the joint before the pool is refilled, and schedule around that.

7 Jurisdiction package

CHIEF ENGINEER AHJ

Send these five items together. Keep the letter short. The summary sheet and the listing do the work.

1. **Covering letter** (template below) on the sender's letterhead
2. **Technical summary** (one page, below)
3. **IAPMO R&T Product Listing**, File No. W-12547, both pages
4. **Product Specification Guide 2026**, specification and installation pages
5. **Photographs** of the property's pump room with the cut zone marked, plus one completed installation elsewhere

Covering letter

TEMPLATE

`[Date]`

`[Name, Title]`
`[Department]`
`[City or County]`

Re: Permit determination for an in-line water conditioning device at `[Property name and address]`

Thank you for your time by phone on `[date]`. This letter confirms our conversation and requests a written determination on whether a permit, plan review, or notification is required for the work described below.

The device. The AquaRev hydrodynamic cavitation device (listed under the brand name "HDC Products") is a passive water conditioning device for swimming pools and spas, certified to NSF/ANSI/CAN 50-2024 by IAPMO Research and Testing, Inc., File No. W-12547, and listed in compliance with the Uniform Swimming Pool, Spa and Hot Tub Code. It is made in the USA of PVC, has no moving parts, no consumable items, no screens or filters, requires no electricity, and has no connection to the water supply, drain, or sewer. Its certified performance claim is a 41% reduction in the chlorine needed to maintain the test pool's water chemistry. It is covered by U.S. Patent No. 10,934,180 and related patents. The listing and specification sheet are enclosed.

The work. At `[Property]` we propose to install `[number]` `[size]`-inch devices, one on each of `[number]` existing, separate bodies of water, each with its own pump and filter. On each system the device is placed on the existing `[size]`-inch PVC return line downstream of the pump, filter, heater,

and chemical feeders. Installation consists of cutting the existing return pipe, solvent-welding the halves of the supplied Schedule 40 PVC unions to the cut ends, and setting the device into the unions. No electrical, gas, water-supply, or drain connection is made. The pump, filter, disinfection equipment, controllers, drains, and suction-side plumbing are not modified. The work will be performed by `[licensed contractor and license number]`.

Enclosed: IAPMO R&T Product Listing (2 pp.), Product Specification Guide excerpt (2 pp.), one-page technical summary, photographs.

We would be grateful for confirmation by email that no permit is required or, if one is required, the permit type, fee, and any plan-check requirement. Technical questions may be directed to AquaRev Water at water@aquarevwater.us or (832) 979-6758.

Sincerely,
`[Name, Title]`
`[Property or AquaRev Water]`
`[Phone · Email]`

Technical summary for authorities having jurisdiction

AquaRev Hydrodynamic Cavitation Device

One page · Rev 1.1 · Sept 2026

PRODUCT TYPE	Water Conditioning Device for pools, spas, and hot tubs. Passive, in-line, no moving parts.
LISTING	IAPMO R&T File No. W-12547. NSF/ANSI/CAN 50-2024. USPC® mark. Uniform Swimming Pool, Spa and Hot Tub Code. Listee: KM LLC dba KD Enterprises, 4348 Waialae Ave. #621, Honolulu, HI 96816. Verify at pld.iapmo.org.
MATERIALS	PVC. NSF/ANSI 372 lead-free. California Prop 65 compliant. Made in Philadelphia, PA.
UTILITIES	None. No electrical, gas, water-supply, drain, sewer, or backwash connection.
FUNCTION	Water passing through fixed internal geometry undergoes hydrodynamic cavitation. Supplements the pool's disinfection; does not replace it. Sanitizer and pH requirements are unchanged.
LOCATION	Return line, after pump, filter, heater, and chemical feeders; before the pool return inlets. Any orientation.
CONNECTION	In-series with two supplied Schedule 40 PVC unions, solvent-welded to the existing pipe. No bypass.
HYDRAULICS	Rated head loss 2.6 psi. Sizes for 2, 3, 4, and 6 in. return lines; 44 to 1,100 gpm by model.
NOT MODIFIED	Pump, filter, heater, disinfection and control equipment, drains and covers, suction plumbing, shell, deck, barriers.
MAINTENANCE	None scheduled. Unions allow removal for inspection. Lifetime limited warranty.
CONTACT	AquaRev Water · water@aquarevwater.us · (832) 979-6758 · aquarevwater.us/data

Determination record

PROPERTY / BODIES OF WATER	
JURISDICTION / DEPARTMENT	
CONTACT NAME / TITLE / PHONE	
CALL DATE / PACKAGE EMAILED	
DETERMINATION	☐ No permit required ☐ Notification only ☐ Permit required: type _____ fee _____
CODE SECTION CITED	
WRITTEN REPLY ON FILE	☐ Yes ☐ Requested, pending
RECORDED BY / DATE	

8 Installation

CONTRACTOR | CHIEF ENGINEER

Expanded from the three-step installation guide in the Product Specification Guide. Where the union or cement manufacturer's instructions are more specific, follow them.

Bring

- [ ] **The device, correct model**
  Match the carton label to the assessment record. Both union halves and O-rings inside.
- [ ] **PVC primer and cement**
  Rated for the pipe size. Heavy-body cement for 4 in. and up. Check the shelf date.
- [ ] **Spare pipe, couplings, elbows**
  Same size and schedule as the return line, for reroutes and recovery.
- [ ] **Pipe cutter or saw, deburring tool**
  A clean square cut is what makes a union seal.
- [ ] **Lockout kit and tags**
  For the pump breaker and every chemical feeder.
- [ ] **Gloves, eye protection, ventilation**
  Solvent cement in a closed pump room needs a fan or an open door.

Before switching anything off

1. Photograph the carton label. Record the model and serial number in the project file.
2. Find the IN and OUT markings. Decide which end faces the pump.
3. Dry-fit the union halves to the device and measure socket bottom to socket bottom. That is your cut length. Write it on the pipe.

> **LOCKOUT FIRST**
> Switch off and lock out the pump at its disconnect, and every chemical feeder on the loop: chlorinator or salt cell, acid or CO_2 feed, and the controller's dosing outputs. Stopping flow with feeders live can concentrate chemicals in the line and off-gas when the pipe is opened.

Procedure

01 Isolate and depressurize

Close return-line isolation valves if fitted; otherwise open the filter drain and let the line empty. Open the filter air relief and confirm the gauge reads zero. Put a bucket under the cut zone.

02 Mark and cut

Mark the cut length on the return pipe, after the last treatment component and as close to the pool return as the room allows. Cut square, remove the section, deburr both ends inside and out, and wipe them clean and dry.

03 Weld the union tails

Prime the pipe end and the union socket, cement both, push fully home with a quarter turn, and hold for 30 seconds. Repeat on the other end. Orient both union nuts so they can be turned by hand once the device is in.

04 Set the device

Check the IN arrow points toward the pump. Seat the device into the union tails with the O-rings in place and the faces clean. Tighten each nut hand-tight plus a small snug. No wrench; PVC unions seal on the O-ring, not on torque. Support the body on a long horizontal run.

05 Cure

Wait the cement manufacturer's cure time for the pipe size and temperature before pressurizing. Two-inch joints at room temperature typically want 1 to 2 hours; larger sizes and cool, damp rooms want longer. When in doubt, wait longer. Use the time to close drains and refill the filter.

06 Restart and check

Open isolation valves and remove the pump lockout only. Start the pump with the air relief open until water runs clear. Inspect both unions and both welds under pressure; a weep at a union means reseat, not tighten harder. Read the

filter gauge; expect up to 2.6 psi more. Then restore the feeders and controller, and photograph the finished install with the listing mark visible.

Installation record

BODY OF WATER / DEVICE SERIAL	
CUT LENGTH USED (IN.)	
CURE TIME ALLOWED	
FILTER GAUGE BEFORE / AFTER (PSI)	
LEAK CHECK AFTER 24 H	☐ Dry ☐ Weep, corrected
INSTALLER / LICENSE / DATE	

9 After installation

CHIEF ENGINEER | MANAGEMENT

The device starts working on the first pump cycle. The water chemistry shifts over the following weeks as combined chlorine, stabilizer, and fine particles come down. Keep dosing exactly as the health code requires and let the controller find the new, lower demand on its own.

The first weeks

- **Expect the filter to load faster in week one.** Agglomerated fines reach the filter for the first time. A backwash or cartridge clean is normal.
- **Do not change controller setpoints.** Reduced demand shows up as less dosing, not as a reason to run leaner than the code allows.
- **Chemistry stays inside the regulated range.** Balanced water is also a warranty condition.

If something looks wrong

SYMPTOM	WHAT TO DO
Filter gauge climbing	Clean the filter first; that is the cause far more often than the device. If the gauge stays high with a clean filter, inspect the device.
Device suspected blocked	Lock out pump and feeders. Loosen the unions, remove the device, turn it 180°, run water backward through it for 5 minutes, and reinstall in the original direction. Then fix whatever let debris past the filter.
Weep at a union	Depressurize, loosen, clean the faces, reseat the O-ring, retighten snug by hand.
Winter	Winterize with the rest of the loop. The device drains through the unions.

Warranty in brief

Lifetime limited warranty on the device body. Unions, O-rings, and external piping are excluded. Installation must be by a qualified professional following AquaRev's documentation, and water must be kept balanced within industry

ranges. Claims go to water@aquarevwater.us with the install date and serial number. Full terms at aquarevwater.us/warranty.

Keep in the project file

- [ ] **Assessment record** for each body of water, with photos
- [ ] **Determination record** with the authority's written reply, or the permit and inspection sign-off
- [ ] **IAPMO listing, spec guide, and warranty terms** as issued at purchase
- [ ] **Installation record** with serial numbers and installer license

References and contacts

DOCUMENT	WHERE
IAPMO R&T Product Listing, File No. W-12547	aquarevwater.us/data · verify at pld.iapmo.org
Product Specification Guide 2026	aquarevwater.us/data
Technical White Paper, April 2026	aquarevwater.us/techpaper
Lab reports and case studies	aquarevwater.us/data
Warranty terms	aquarevwater.us/warranty
Sales and AHJ package support	water@aquarevwater.us · (832) 979-6758
Technical support and warranty claims	aquarev@hdcproductsinc.com · (808) 628-0208

Performance figures are from independent laboratory studies. Individual results vary. Permit determinations rest with the authority having jurisdiction.